

Telecom Customer Segmentation

Model Implentation, Cluster Analysis and Customer Profiling

Executive Summary

This report presents the results of a K-Means customer segmentation analysis conducted on a telecom customer base. The objective was to identify customer groups with similar characteristics and translate the resulting clusters into meaningful business segments. Four clusters were selected using the Elbow Method and subsequently examined across three primary dimensions: household structure, customer tenure, and monthly charges. The analysis reveals clear differences in the composition and observed churn behavior of the four clusters. The most significant distinction is lifecycle-related: one cluster consists entirely of short-tenure customers and records the highest observed churn rate of 51.0%, while another consists entirely of long-tenure customers and records the lowest observed churn rate of 11.1%. Household structure and spending level provide additional differentiation, particularly between customers with and without dependents and between lower- and higher-value accounts. Based on these patterns, the clusters were translated into four actionable customer segments: **New Independent Customers, Family Value Customers, Loyal Premium Customers, and Mid-Tenured Independent Customers**. The segmentation provides a practical foundation for differentiated retention, customer-value, and relationship-management strategies.

Cluster Analysis and Modelling

Clustering Approach

K-Means clustering was applied to identify groups of customers with similar characteristics. The model used the selected customer attributes following the exploratory analysis and feature-engineering stages. The clustering variables represented three important dimensions of the customer relationship:

Dependents — household structure.

Tenure Groups — customer lifecycle stage.

Monthly Charges Groups — customer spending level.

The model was trained using the training dataset, after which the cluster assignments were linked back to the original customer records for interpretation. The purpose of this stage was not simply to produce four statistical groups, but to determine whether those groups represented meaningful and actionable differences in customer behavior.

Model Optimization

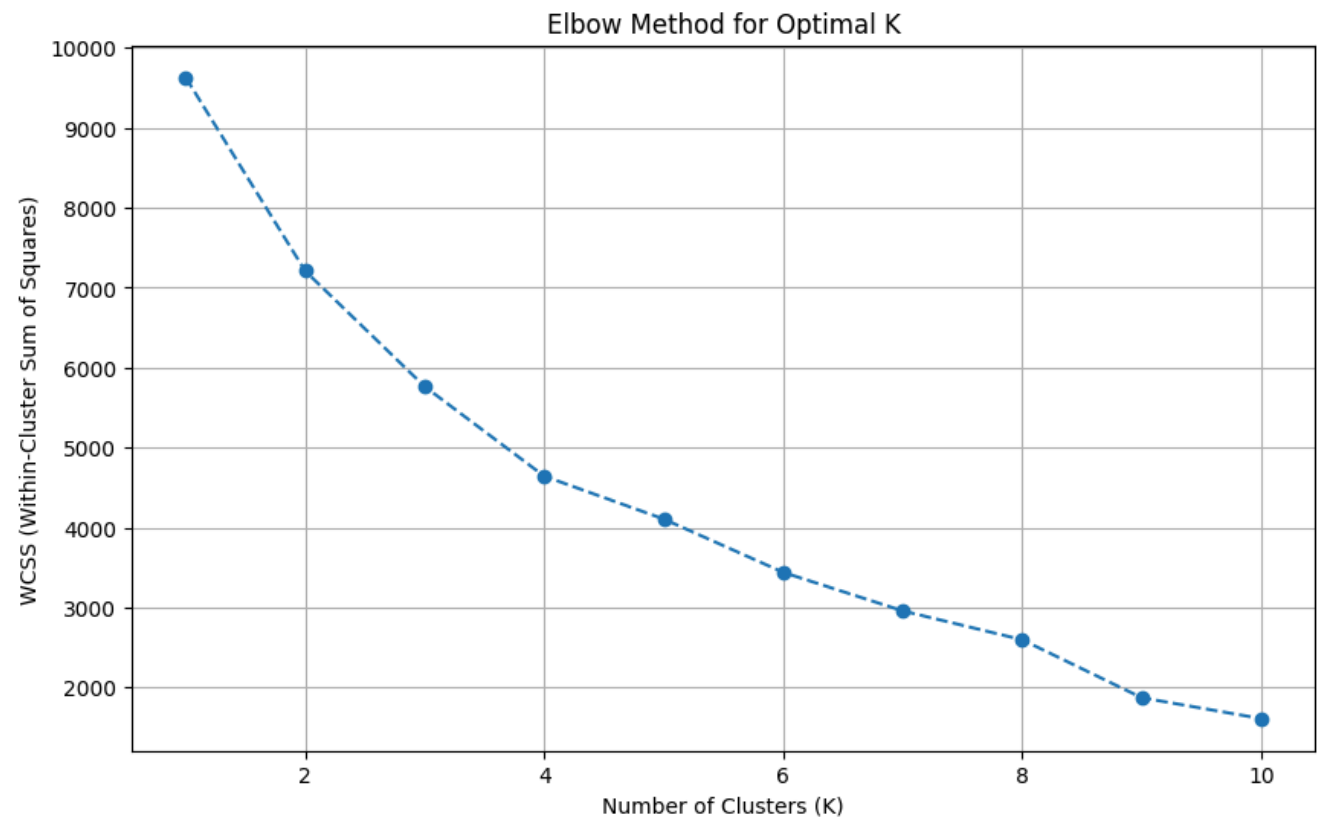
Determining the Number of Clusters

The Elbow Method was used to determine the appropriate number of clusters. The WCSS curve in Figure 1 shows substantial reductions in within-cluster variation as the number of clusters increases from $K=1$ through $K=3$. The rate of improvement then begins to decline around $K=4$, indicating that additional clusters yield progressively smaller improvements in explaining variation in the data. Based on this point of diminishing returns, $K=4$ was selected as the optimal number of clusters.

The four-cluster solution was also considered appropriate from a business perspective because it produced groups with clear differences in customer lifecycle, household structure, and spending behavior.

Figure 1

Elbow Method



Overall Cluster Comparison

Table 1**Churn and Retention Rates Across Clusters**

Cluster	Observed Churn	Observed Retention
Cluster 0	51.0%	49.0%
Cluster 3	27.3%	72.7%
Cluster 1	17.1%	82.9%
Cluster 2	11.1%	88.9%

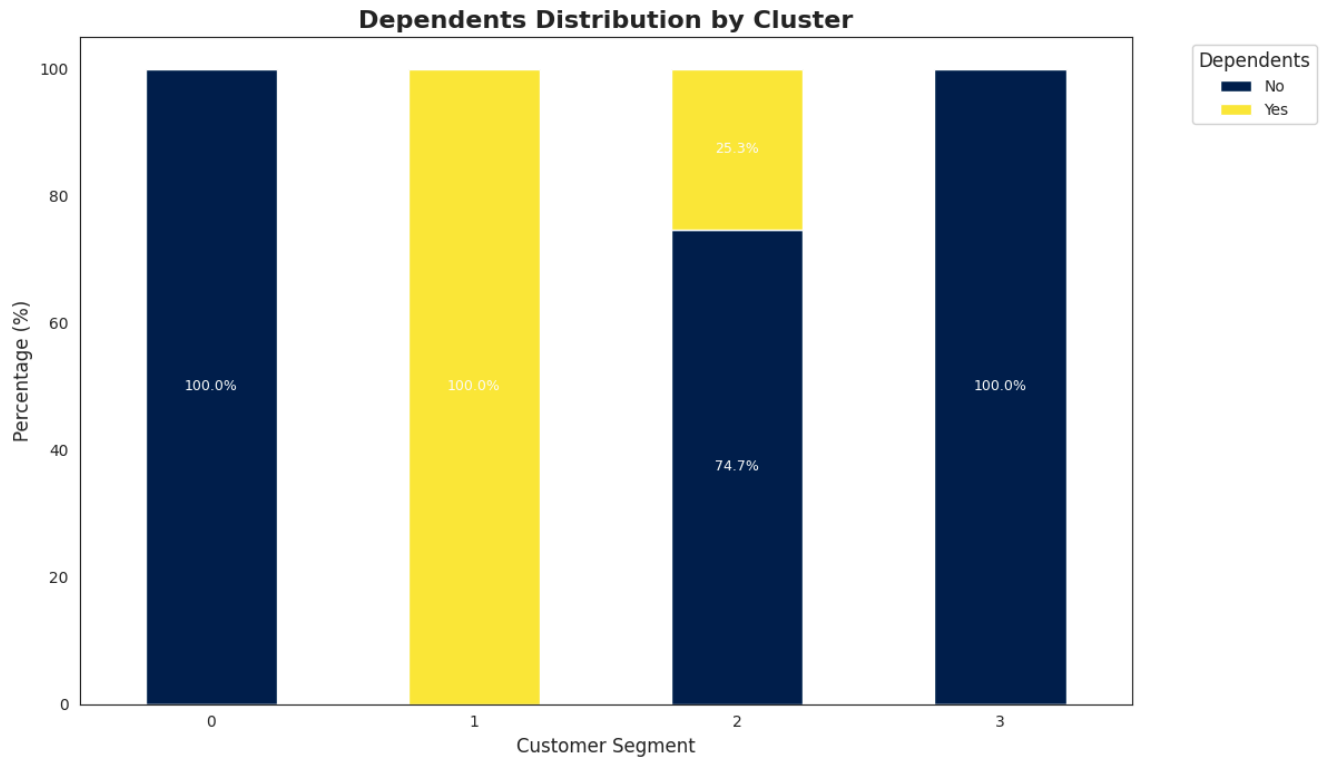
The clusters demonstrate substantial differences in observed churn. Cluster 0 has the highest observed churn at 51.0%, meaning that more than half of the customers in this group left. Cluster 3 has the second-highest churn rate at 27.3%. Cluster 1 and Cluster 2 are comparatively stable, with observed churn rates of 17.1% and 11.1%, respectively. This establishes an important difference between the clusters, but churn alone does not explain why the groups differ. The next sections, therefore, examine the individual dimensions that characterise each cluster.

Cluster Analysis by Customer Dimensions

Household Structure (Dependents)

Figure 2

Dependents Distribution Across Clusters

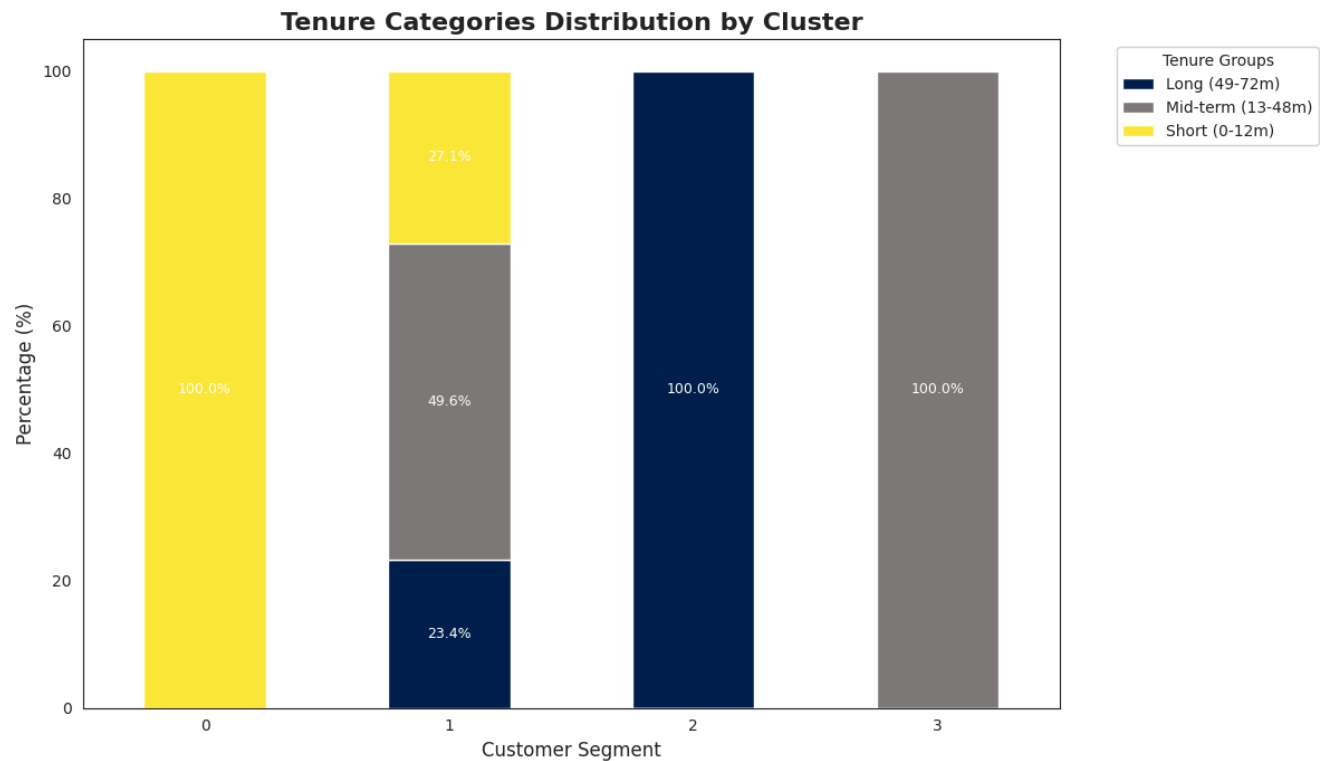


Household structure provides one of the clearest distinctions between the clusters. Cluster 0 consists entirely of customers without dependents. Cluster 3 is also entirely composed of customers without dependents. In contrast, Cluster 1 consists entirely of customers with dependents. Cluster 2 is more diverse, with customers without dependents making up the majority at **74.7%**, compared with 25.3% with dependents. This indicates that the presence or absence of dependents is a major characteristic distinguishing the customer groups.

Customer Tenure

Figure 3

Tenure Distribution Across Clusters



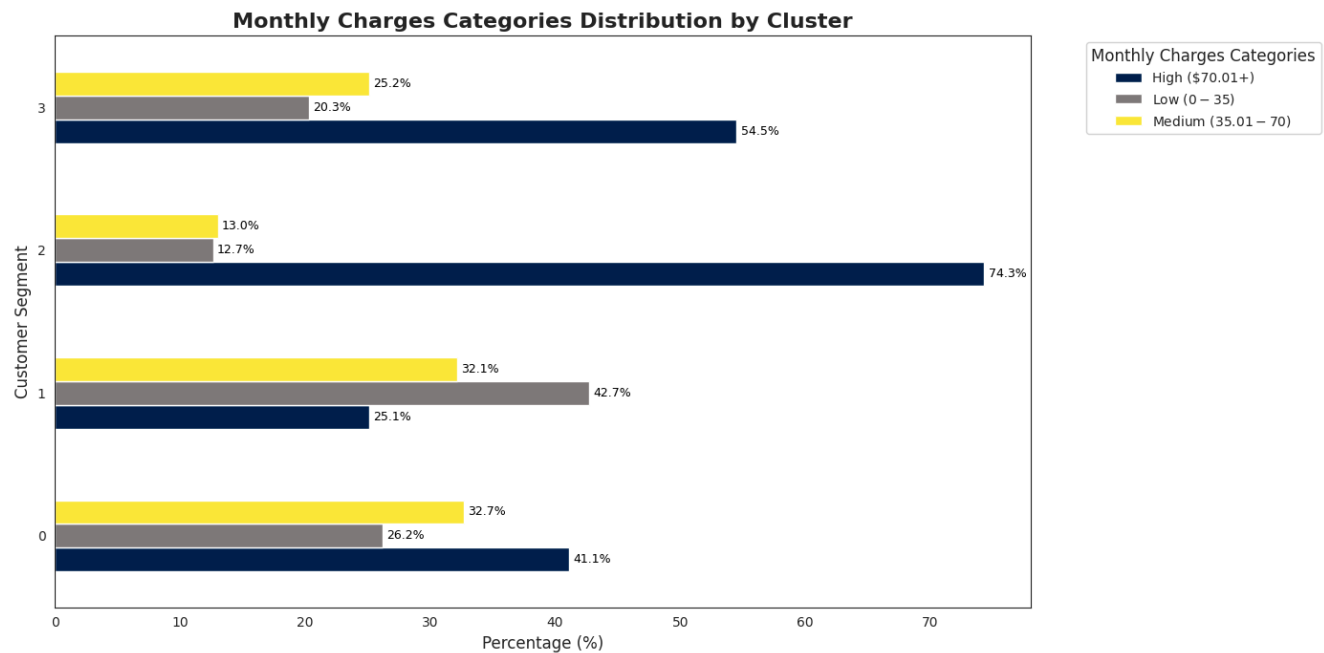
Tenure provides the strongest lifecycle differentiation between the clusters. This dimension creates a clear progression across three of the clusters. Cluster 0 represents customers at the earliest stage of the relationship, Cluster 3 represents customers who have progressed to the middle stage, and Cluster 2 represents customers who have maintained the relationship the longest. Cluster 1 differs because its customers are distributed across the entire customer lifecycle.

This dimension is particularly important because the clusters with different tenure profiles also demonstrate substantially different observed churn rates. The short-tenure Cluster 0 records the highest churn, while the long-tenure Cluster 2 records the lowest. However, this relationship should be interpreted as an association rather than evidence that tenure itself causes churn.

Monthly Charges

Figure 4

Monthly Charges Distribution Across Clusters



Monthly charges provide a further distinction between the customer groups. Cluster 2 has the highest concentration of high-paying customers, with 74.3% paying more than \$70 per month. Cluster 3 also has a strong concentration of high-paying customers, with 54.5% in the highest charge category. Cluster 0 has a more balanced distribution, while Cluster 1 is more concentrated toward lower and medium charges. An important business observation is that high monthly charges do not necessarily correlate with high churn.

Cluster 2 has the highest proportion of high-paying customers but also has the lowest observed churn rate. This demonstrates why customer spending should be interpreted alongside other dimensions, such as tenure and household structure.

Integrating the Dimensions

The previous analysis shows that no single dimension fully describes the customer groups. The value of the clustering exercise comes from examining the dimensions together.

Table 2

Key Cluster Characteristics

Cluster	Dependents	Tenure	Monthly Charges	Churn Rate	Key Characteristics
Cluster 0	100% No	100% Short	Mixed	51%	New independent customers with high risk of churn
Cluster 3	100% No	100% Mid	Majority High	27.3%	Independent established high value customers with moderate risk of churn
Cluster 1	100% Yes	All	Low and Medium	17.1%	Family oriented and value conscious customers with a low risk of churn
Cluster 2	74.7% No	100% Long	74.3% High	11.1%	Long term high value customers with very low risk of churn.

Customer Segment Profiles

Cluster 0: New Independent Customers

Cluster 0 represents customers who are at the beginning of their relationship with the telecom provider. All customers in this group have short tenure and no dependents. The segment records the highest observed churn rate at 51.0%. The combination of early lifecycle stage and independent household structure makes this group particularly important for customer onboarding and early retention. Strengthening relationships during the first year, conducting proactive customer check-ins, conducting surveys to investigate whether high monthly charges are creating perceived value concerns, and monitoring installations and technical issues would reduce the churn rate in the early stages.

Cluster 1: Family Value Customers

Cluster 1 consists entirely of customers with dependents and is distributed across all three tenure categories. The majority of customers fall within the low-to-medium monthly charges. The segment has an observed churn rate of 17.1%, making it relatively stable compared with the higher-risk clusters.

The combination of household structure, moderate spending, and relatively low churn supports its classification as a family-oriented value segment. The business priority for this segment is to protect retention while increasing household value. Recommended actions include offering relevant parental-control or smart-home solutions and exploring multiline and household services.

Cluster 2: Loyal Premium Customers

Cluster 2 is the most stable and commercially attractive segment. All customers have long tenure, and 74.3% pay more than \$70 per month. The segment records the lowest observed churn rate at 11.1%. The combination of long-term customer relationships, high monthly spending, and low observed churn makes this group valuable. Offering premium service benefits, providing priority customer service, and implementing a loyalty recognition program would help protect loyalty and maximise long-term customer value for this segment.

Cluster 3: Mid-Tenured Independent Customers

Cluster 3 consists entirely of customers without dependents and entirely of customers with mid-term tenure. More than half of the customers in this segment pay above \$70 per month, making it a relatively high-value group. However, the segment has an observed churn rate of 27.3%, considerably higher than the two more stable segments. Introducing targeted loyalty benefits, conducting proactive renewal engagements, offering value-added services, and monitoring customers approaching key renewal periods would strengthen retention and perceived value during the mid-customer lifecycle.

Business Insights

The dimensional analysis and resulting segment profiles produce several important business insights.

1. Customer lifecycle is a major differentiator

The strongest structural distinction between the clusters is tenure. The customer base shows a clear progression from short-tenured customers in Cluster 0 to mid-tenured customers in Cluster 3 and long-tenured customers in Cluster 2. This suggests that customer management strategies should consider the stage of the customer relationship, rather than applying the same retention strategy to every customer.

2. Household structure separates important customer groups

Customers with dependents are concentrated in Cluster 1, while Clusters 0 and 3 consist entirely of customers without dependents. This creates an opportunity to differentiate communication and product strategies according to household needs.

3. High-value customers are not necessarily high-risk

Cluster 2 provides an important counterpoint to the assumption that customers paying more are necessarily more likely to churn. Although 74.3% of Cluster 2 customers pay more than \$70, the segment has the lowest observed churn rate at 11.1%. Therefore, monthly charges should not be used in isolation to identify churn risk.

4. The highest-risk customers are concentrated in the early lifecycle

Cluster 0 has an observed churn rate of 51.0% and consists entirely of customers with tenure below 12 months. This makes early customer experience a major area for management attention. The organisation should therefore prioritise understanding what happens in the first year of the customer relationship.

5. Different segments require different strategies

The analysis demonstrates why a single customer-retention strategy is unlikely to be optimal.

Conclusion

The analysis identifies four distinct and actionable customer segments by examining the combined dimensions of household structure, customer tenure, and monthly charges. The dimensional analysis reveals a clear lifecycle pattern, with short-tenured customers exhibiting the highest observed churn, long-tenured customers demonstrating the strongest retention, and the highest concentration of premium spending. These characteristics support the development of four meaningful business segments: New Independent Customers, Family Value Customers, Loyal Premium Customers, and Mid-Tenure Independent Customers. The segmentation provides management with a more structured understanding of the customer base and demonstrates that different groups require different business strategies. The immediate priority should be reducing early churn among New Independent Customers, while Mid-Tenure Independent Customers require targeted value and loyalty interventions. At the same time, Family Value Customers should be protected and developed through relevant household offerings, while Loyal Premium Customers should be retained and encouraged to become long-term advocates. Overall, the analysis provides a practical foundation for moving from broad customer management toward targeted, segment-specific decision-making.